Daniel Teal PhD

Mechanical, Electrical, and Semiconductor Process Engineering

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I'm looking for interesting engineering work in advanced manufacturing.

Experience

2019–now **University of California, Berkeley, electrical engineering PhD student**

Studied microfabrication. Developed direct-write nanoparticle deposition techniques. Built custom vacuum chambers, electronics, and process equipment. Also built world's longest MEMS electrostatic motor in Berkeley Marvell NanoLab research fab. Advised by Prof. Kristofer Pister.

2024–now **Microfabrication Startup (in stealth), founder**

Building on my PhD work to provide fast custom semiconductor manufacturing services via direct-write printing. Currently bootstrapping, part-time/hobby-level.

8/22–5/24 **Berkeley EE143 Microfabrication Class, lead teaching assistant**

Taught hands-on lab course (Berkeley EE143) building NMOS transistors from scratch on silicon wafers, four semesters in a row. Instructed students, maintained equipment, troubleshot processes, and managed other TAs.

8/17–6/19 Fan Nanomaterial Innovation Lab, research assistant. Built real-time electrostatic control system.

5/18–8/18 Nano Functionality Integration Group, Japan, research intern. Studied nanowire networks.

6/17–8/17 Cornell NanoScale Facility, research intern. Learned microfabrication, built and tested SAW MEMS.

9/15–5/19 UT Maker Studios, student assistant. Trained students on & maintained 3D printers and laser cutters.

9/15–5/19 UT Robotics & Automation Society, member. Built assorted robots including motorized couch.

8/17–5/19 UT ME Undergraduate Advisory Board, member. Ran the first UT Createathon hardware hackathon.

11/16–5/17 Zheng Research Group, research assistant. Reviewed plasmonic nanostructures.

11/15–11/16 UT Advanced Manufacturing Center, research assistant. Built high-speed impact test chamber.

6/15–8/15 UT Applied Research Laboratories, Intern. Acoustically detected quadcopters.

9/11–6/15 FIRST Tech Challenge, Design & Build Lead, LASA High School Robotics. 2014-2015 world finalists.

6/11–8/15 DIY 3D printer design & construction, hobbyist. Self-taught mechatronics.

Education

University of California, Berkeley
PhD, Electrical Engineering and Computer Science, 2024
microfabrication, devices, process engineering, MEMS

University of Texas at Austin
BS Mechanical Engineering, BS Mathematics, 2019
solids/dynamics, material science, thermodynamics, organic chemistry, heat transfer, numerical analysis

Skills

Microfabrication: MEMS, litho, deposition, etching, metrology, packaging
Hardware: automation, thermal systems, vacuum systems
Prototyping: 3D printing, laser cutting, machining
CAD & FEA: SolidWorks, Fusion 360, KiCad
Programming: Python, C/C++, Rust, Linux, LaTeX, HTML/CSS

Portfolio
dteal.org/portfolio.html



2024